

environment, making it a versatile tool for various use cases.

- *Community-driven signatures:* The open source nature of Snort means that it benefits from a large community that actively contributes to updating and improving attack signatures.

Kali Linux: The ultimate offensive security platform

Kali Linux is a Debian-based Linux distribution designed for penetration testing and digital forensics. It comes preloaded with hundreds of tools for security testing, vulnerability scanning, and exploitation, making it the go-to OS for cybersecurity professionals.

Why Kali Linux remains essential for security testing in 2025

- *Comprehensive toolset:* Kali Linux includes tools for network scanning (Nmap), vulnerability scanning (OpenVAS), password cracking (John the Ripper), and more, providing a one-stop platform for penetration testers.
- *Live boot options:* It can be run from a live USB or bootable disk, making it easy for security professionals to conduct assessments on any system without leaving traces.
- *Active community and updates:* Kali Linux is maintained by Offensive Security, ensuring that the tools included are up to date with the latest threats and vulnerabilities.

Cloud management: Open source tools to manage multi-cloud environments

With the increasing adoption of cloud services, many organisations are embracing multi-cloud strategies to avoid vendor lock-in, improve resilience, and optimise costs. Multi-cloud environments allow companies to leverage the best features and services from multiple cloud providers like AWS, Azure, and Google Cloud. However, managing these diverse environments can be complex and requires effective orchestration and monitoring tools.

Here are some of the top open source tools that play a crucial role in managing multi-cloud deployments.

OpenStack: The leading open source cloud platform

OpenStack is a cloud computing platform that allows organisations to create and manage public and private clouds. It provides a wide range of services, including compute (via Nova), storage (via Swift), networking (via Neutron), and identity management (via Keystone). OpenStack is known for its flexibility and scalability, making it ideal for managing multi-cloud environments.

Why OpenStack is essential for multi-cloud management in 2025

- *Flexibility:* OpenStack allows organisations to customise their cloud infrastructure based on specific needs, providing a high degree of control over how resources are provisioned and managed.
- *Vendor agnostic:* OpenStack can be deployed across different cloud platforms (public, private, and hybrid clouds), making it an effective tool for multi-cloud management.
- *Comprehensive ecosystem:* OpenStack offers a full suite of tools for managing virtualised resources, storage, networking, and compute resources, streamlining the deployment and operation of cloud environments.
- *Active community support:* With its large, vibrant community, OpenStack constantly evolves to incorporate the latest cloud technologies and security features, making it a long-term solution for cloud management.

Prometheus and Grafana: Monitoring and observability for multi-cloud environments

Prometheus is an open source monitoring and alerting toolkit designed for reliability and scalability in complex systems like multi-cloud environments. It collects and stores metrics from various cloud services and applications, providing a detailed view of system performance. Grafana is a visualisation tool that works in tandem with Prometheus to provide interactive dashboards and real-time metrics.

Why Prometheus and Grafana are key for cloud management in 2025

- *Real-time monitoring:* Prometheus provides real-time monitoring of cloud infrastructure, collecting data about CPU usage, memory, disk space, network traffic, and application performance, ensuring cloud resources are being utilised efficiently.
- *Custom dashboards:* With Grafana, users can create custom dashboards that display critical performance data from multiple cloud platforms, making it easier to monitor the health of a multi-cloud infrastructure.
- *Alerts and notifications:* Prometheus' alerting capabilities ensure that teams are immediately notified of performance issues, resource shortages, or any critical failures in the cloud environment, allowing for swift remediation.
- *Cloud-native:* Prometheus is particularly well-suited for cloud-native environments, and with integration support for Kubernetes, it helps manage dynamic, containerised cloud applications seamlessly.